

ROBOTICS CAN BE A GREAT FUN AT WORK

If you are an electronics engineer comfortable with both mechanics and computer programming and having a drive to build new things, consider a career in robotics



Robots assembling vehicle floors

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Let me take you to a classroom of an elementary school in Daegu, southeast of Seoul: Students are working at their desks as the teacher reads books to them and calls out instructions. It's a typical classroom scene except that the teacher, about 1-metre tall, is made of plastic and computer circuits.

Confused? Well, it's an English language class taught by a robot teacher that rolls around the classroom on wheels, asking questions in English and dancing to music. The egg-shaped robot called 'Engkey' is linked to and controlled remotely by an English

teacher outside the classroom (located as far away as Australia) whose face appears on the screen of the robot. Besides being popular with children, Engkey is a cost-effective way to help address South Korea's shortage of qualified native-English speaking teachers.

Robotics, which has grown far beyond the realms of science fictions, has always been a fascinating subject. From simple robot toys to industrial robotics, robotics technology is today firmly entrenched into our day-to-day life and impacts our way of life much more than we think. The advancement in recent technologies continues to bring cyber and physical worlds to-

gether, enabling the robotics industry to grow even faster.

Let's explore a career in robotics engineering.

Know the field

Let us first define the terms 'robot' and 'robotics.' According to the Robot Institute of America (1979), a robot is "A reprogrammable, multifunctional manipulator designed to move material, parts, tools or specialised devices through various programmed motions for the performance of a variety of tasks."

Webster defines a robot as "An automatic device that performs functions normally ascribed to humans or a machine in the form of a human." More generally, it is a machine that functions in place of a living person.

Now coming to robotics, it's a multidisciplinary area which integrates intelligent control, communications, computer vision, mechatronics, sensor fusion, design and many other aspects on a single platform for a better service. This branch of technology is related to the sciences of electronics, engineering, mechanics and software.

"Robotics is a highly technical field of advanced engineering and technology. It involves practical skills backed by higher-end theory. It calls for innovative thinking, application and creativity—skills which are not found in textbooks," explains Dr G. Venkatesh, adjunct professor, Global Academy of Technology, and managing director, Bangalore Robotics.

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